Scalars & Vectors

Exam code: 9MA0

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Scalars & vectors

Mechanics uses a variety of different scalar and vector quantities.

What are scalars?

- **Scalars** are quantities without direction, they only have a **size (magnitude)**
  - e.g. a **speed** of 7 m/s, a **distance** of 5 m, a **time** of 15 s or a **mass** of 12 kg
- Scalars can only be **positive**- you can't have a negative distance or speed!

What are vectors?

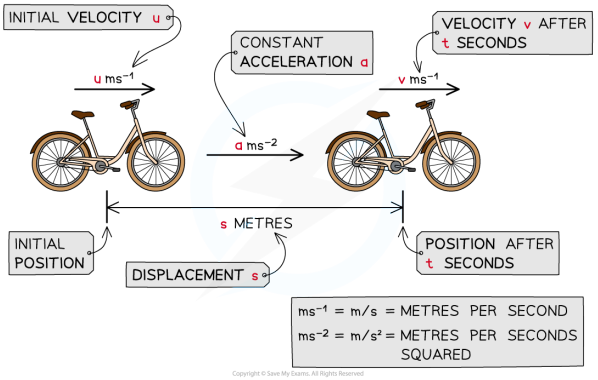
- **Vectors** are quantities which also have a **direction**, this is what makes them more than just a scalar
  - e.g. two objects with **velocities** of 7 m/s and -7 m/s are travelling at the **same speed** but in **opposite directions**
- Vector quantities can have **positive** or **negative components**
- Some examples of vector quantities you may come across are **displacement**, **velocity**, **acceleration**, **force/weight**, **momentum**
  - **Displacement** is the distance moved in a given direction from a starting point

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Vector quantities of displacement, velocity and acceleration

Worked Example

State whether each of the following is a scalar or a vector quantity.

a) A speed boat travels at 3m/s on a bearing of 052°

b) A garden is 1.7 m wide

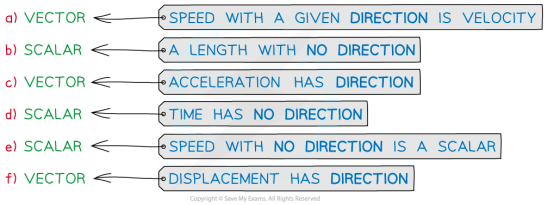
c) A car accelerates at 5.4 ms^{-2}

d) A film lasts 2 hours 17 minutes

e) An athlete runs at an average speed of 10.44 ms^{-1}

f) A ball rolls forwards 60 cm before stopping

Answer:



Examiner Tips and Tricks

- Make sure you fully understand the definitions of all the words in this section so that you can be clear about what your exam question is asking of you
- Vectors appear in **bold** (non-italic) font in textbooks, on exam papers, etc (i.e. ***F***, ***a***) but in handwriting should be underlined (i.e. $\underline{F}$, $\underline{a}$)

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